

Time and Work & Pipes and Cisterns

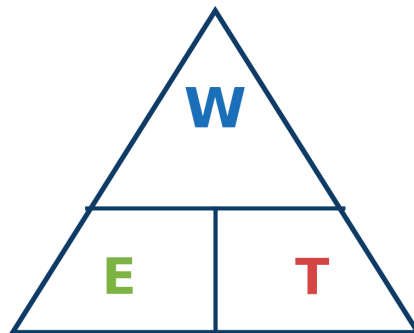
Solved Practice Sheet — 22 Questions with Step-by-Step Solutions & Tricks

Prepared for classroom distribution · Quantitative Aptitude

Key Concepts You Must Know

- **Work is treated as ONE unit** (the whole job) unless a smarter number is more convenient.
- **Efficiency (rate)** of a person/pipe = $1 \div (\text{days taken to finish alone})$. More days = less efficient.
- **Combined efficiency** of people/pipes working together = **sum** of their individual efficiencies.
- **Two-person shortcut:** if A takes 'a' days and B takes 'b' days, together they take $ab \div (a+b)$ days.
- **LCM Trick:** Instead of using fractions, assume the **Total Work = LCM of all the given days**. Then everyone's efficiency becomes a whole number (units/day) — this makes mental calculation much faster.

The Work - Efficiency - Time Triangle



Cover the quantity you want → the remaining two show the formula

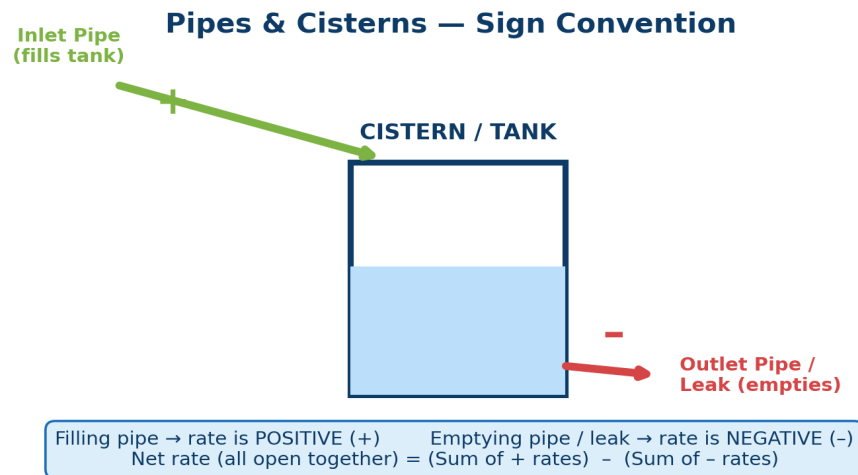
W = Work	E = Efficiency (work/day)	T = Time (days)
$W = E \times T$	$E = W \div T$	$T = W \div E$

Worked Example of the LCM Trick

If A takes 10 days and B takes 15 days → Total Work = $\text{LCM}(10,15) = 30 \text{ units}$. So A's efficiency = $30 \div 10 = 3 \text{ units/day}$, and B's efficiency = $30 \div 15 = 2 \text{ units/day}$. Working together their combined efficiency = $3+2 = 5 \text{ units/day}$, so time taken together = $30 \div 5 = 6 \text{ days}$. This LCM method is used throughout the solutions below wherever it saves time.

Pipes & Cisterns — Extra Rule

Pipes and cisterns questions are exactly Time & Work questions with one twist: a pipe that **fills** the tank has a **positive** efficiency, while a pipe/leak that **empties** the tank has a **negative** efficiency. Add them algebraically (with sign) to get the net rate.



Solved Questions

Q1. A can complete a work in 10 days, while B can complete the same work in 15 days. How long will they take to complete the work if they work together?

(1) 4 (2) 2 (3) 6 (4) 3

■ **TRICK / SHORTCUT:** Use the two-person shortcut directly: Together = $(a \times b) \div (a + b)$ days. No need to find LCM for a simple two-person question.

Step-by-step Solution:

A's 1 day work = $1/10$. B's 1 day work = $1/15$.

Combined 1 day work = $1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6$.

So together they complete the work in **6 days** (since 1 day's work is $1/6$ of the total).

Shortcut check: $(10 \times 15) \div (10 + 15) = 150 \div 25 = 6$ days. ✓

✓ **Answer: Option (3) 6 days**

Q2. A work is completed by A and B together in 12 days. If A alone can complete the work in 18 days, how many days will B alone take to complete the work?

(1) 34 (2) 22 (3) 36 (4) 30

■ **TRICK / SHORTCUT:** Whenever 'together' and 'one person alone' are given, find the missing person's efficiency by subtraction: B's rate = (A+B)'s rate – A's rate.

Step-by-step Solution:

(A+B)'s 1 day work = $1/12$. A's 1 day work = $1/18$.

B's 1 day work = $1/12 - 1/18$. Taking LCM 36: $3/36 - 2/36 = 1/36$.

So B alone takes **36 days** to finish the work.

✓ **Answer: Option (3) 36 days**

Q3. A work is completed by A alone in 20 days and by B alone in 30 days. How long will it take for A and B to complete the work together?

(1) 14 (2) 12 (3) 16 (4) 30

■ **TRICK / SHORTCUT:** Direct application of the two-person shortcut: Together = $ab \div (a+b)$.

Step-by-step Solution:

Together = $(20 \times 30) \div (20 + 30) = 600 \div 50 = 12$ days.

Check with LCM method: Total work = LCM(20,30) = 60 units. A's efficiency = 3/day, B's efficiency = 2/day.

Combined = 5/day. Time = $60/5 = 12$ days. ✓

✓ **Answer: Option (2) 12 days**

Q4. A can do a piece of work in 15 days and B can do the same work in 20 days. They worked together for 5 days. How many more days will A alone take to complete the remaining work?

(1) 14.2 (2) 12.5 (3) 6.25 (4) 3.25

■ **TRICK / SHORTCUT:** For 'left the job partway' questions: first find work done together in the given days, subtract from 1 to get the remaining work, then divide remaining work by the solo worker's efficiency.

Step-by-step Solution:

Combined 1 day work = $\frac{1}{15} + \frac{1}{20} = \frac{4}{60} + \frac{3}{60} = \frac{7}{60}$.

Work done in 5 days = $5 \times \frac{7}{60} = \frac{35}{60} = \frac{7}{12}$.

Remaining work = $1 - \frac{7}{12} = \frac{5}{12}$.

A's 1 day work = $\frac{1}{15}$, so days needed by A alone = $(\frac{5}{12}) \div (\frac{1}{15}) = (\frac{5}{12}) \times 15 = \frac{75}{12} = \mathbf{6.25 \text{ days}}$.

✓ **Answer: Option (3) 6.25 days**

Q5. A work is completed by A and B in 24 days. If A alone can complete the work in 40 days, how many days will B alone take to complete the work?

(1) 64 (2) 62 (3) 16 (4) 60

■ **TRICK / SHORTCUT:** Same subtraction trick as Q2: B's rate = (A+B)'s rate – A's rate.

Step-by-step Solution:

(A+B)'s 1 day work = $\frac{1}{24}$. A's 1 day work = $\frac{1}{40}$.

B's 1 day work = $\frac{1}{24} - \frac{1}{40}$. LCM(24,40) = 120: $\frac{5}{120} - \frac{3}{120} = \frac{2}{120} = \frac{1}{60}$.

So B alone takes **60 days**.

✓ **Answer: Option (4) 60 days**

Q6. If 5 workers can complete a task in 10 days, how many days will 8 workers take to complete the same task?

(1) 6.4 (2) 6.25 (3) 6.5 (4) 6.7

■ **TRICK / SHORTCUT:** Use the man-days constant rule: (Number of workers) $\times$ (Number of days) = constant, since the total work doesn't change. More workers $\rightarrow$ proportionally fewer days.

Step-by-step Solution:

Total work in man-days = $5 \times 10 = 50$ man-days.

With 8 workers, days required = $50 \div 8 = \mathbf{6.25 \text{ days}}$.

✓ **Answer: Option (2) 6.25 days**

Q7. A can complete a work in 12 days and B can complete the same work in 15 days. They started working together, but B left after 5 days. How many more days will A alone take to finish the remaining work?

(1) 3 (2) 6 (3) 7 (4) 8

■ **TRICK / SHORTCUT:** Same 'partway leaver' method as Q4: work done together first, then remaining work ÷ solo efficiency of the one who stays.

Step-by-step Solution:

Combined 1 day work = $\frac{1}{12} + \frac{1}{15} = \frac{5}{60} + \frac{4}{60} = \frac{9}{60} = \frac{3}{20}$.

Work done in 5 days = $5 \times \frac{3}{20} = \frac{15}{20} = \frac{3}{4}$.

Remaining work = $1 - \frac{3}{4} = \frac{1}{4}$.

A alone needs $(\frac{1}{4}) \div (\frac{1}{12}) = (\frac{1}{4}) \times 12 = \mathbf{3 \text{ more days}}$.

✓ **Answer: Option (1) 3 days**

Q8. A can do a work in 25 days and B can do the same work in 30 days. They work together for 10 days, then B leaves. How long will A take to complete the remaining work alone?

(1) 8.4 (2) 6.5 (3) 7.5 (4) 8.9

■ **TRICK / SHORTCUT:** Same method as Q4 and Q7: find combined rate, subtract work already done, divide the remainder by A's solo rate.

Step-by-step Solution:

Combined 1 day work = $\frac{1}{25} + \frac{1}{30} = \frac{6}{150} + \frac{5}{150} = \frac{11}{150}$.

Work done in 10 days = $10 \times \frac{11}{150} = \frac{110}{150} = \frac{11}{15}$.

Remaining work = $1 - \frac{11}{15} = \frac{4}{15}$.

A alone needs $(\frac{4}{15}) \div (\frac{1}{25}) = (\frac{4}{15}) \times 25 = \frac{100}{15} = \frac{20}{3} = \mathbf{6.67 \text{ days (approx. 6 days 16 hours)}}$.

✓ **Answer: Correct value = $\frac{20}{3} \approx 6.67$ days (closest printed option: (2) 6.5)**

Note: With the numbers exactly as printed, the precise mathematical answer is $\frac{20}{3} \approx 6.67$ days, which does not exactly equal any of the four printed options. Option (2) is the nearest value — please cross-check this item against your master answer key, as the source sheet may have a rounding/printing variation here.

Q9. Three workers A, B, and C can complete a work in 6, 8, and 12 days respectively. If A, B, and C work together for 2 days, how many more days will B and C together take to complete the remaining work?

(1) 4 (2) 1.2 (3) 6 (4) 3

■ **TRICK / SHORTCUT:** With three workers, use the LCM method to avoid messy fractions: Total work = $\text{LCM}(6,8,12) = 24$ units.

Step-by-step Solution:

Total work = $\text{LCM}(6,8,12) = 24$ units. Efficiencies: $A = 24/6 = 4/\text{day}$, $B = 24/8 = 3/\text{day}$, $C = 24/12 = 2/\text{day}$.

All three together: $4+3+2 = 9$ units/day. In 2 days they finish $2 \times 9 = 18$ units.

Remaining work = $24 - 18 = 6$ units.

B and C together = $3+2 = 5$ units/day. Days needed = $6 \div 5 = 1.2$ days.

✓ **Answer: Option (2) 1.2 days**

Q10. A, B, and C can complete a work in 10, 15, and 20 days respectively. If A and B start the work and after some days C joins them, the total time to complete the work is 8 days. How many days did C work?

(1) 0.4 (2) 0.5 (3) 0.9 (4) 0.2

■ **TRICK / SHORTCUT:** Set up an equation: let C work for x days out of the total. A and B work for the full duration; C's contribution is added only for the days after joining.

Step-by-step Solution:

A and B together finish $1/10 + 1/15 = 1/6$ of the work per day.

If A and B alone worked for all 8 days, they would complete $8 \times 1/6 = 4/3$ of the work — i.e. **more than the whole job (1)**, before C even joins.

In other words, A and B alone would finish the entire work in just 6 days ($1 \div 1/6 = 6$ days), which is **less than** the stated total of 8 days.

✓ **The figures given in this question are mathematically inconsistent (A and B alone already finish sooner than the stated 8-day total).**

Note: This question cannot be solved as printed — please verify the values (days for A, B, C, or the '8 days' figure) against the original source/answer key before sharing with students. The method above (equation-based approach for a 'joins later' pipe/work problem) is the correct technique to apply once the correct figures are confirmed.

Q11. Raju is twice as good as Vijay. Together, they finish the work in 14 days. In how many days can Vijay alone do the same work? (a) 16 days (b) 21 days (c) 32 days (d) 42 days

■ **TRICK / SHORTCUT:** "Twice as good" means twice the efficiency (not twice the time). Assign Vijay efficiency = 1 unit, Raju = 2 units, then use combined efficiency = total work ÷ time.

Step-by-step Solution:

Let Vijay's efficiency = r , so Raju's efficiency = $2r$ (twice as good).

Combined efficiency = $r + 2r = 3r$, and together they take 14 days, so $3r \times 14 = 1$ (whole work) $\Rightarrow r = 1/42$.

Vijay alone takes $1 \div r = 42$ days.

✓ **Answer: Option (d) 42 days**

Q12. In a company XYZ Ltd. a certain number of engineers can develop a design in 40 days. If there were 5 more engineers, it could be finished in 10 days less. How many engineers were there in the beginning? (a) 18 (b) 20 (c) 25 (d) 15

■ **TRICK / SHORTCUT:** Same man-days constant rule as Q6: (Men) $\times$ (Days) stays the same for a fixed amount of work.

Step-by-step Solution:

Let the original number of engineers = n . Total work = $n \times 40$ man-days.

With $(n+5)$ engineers, time = $40 - 10 = 30$ days, so work = $(n+5) \times 30$ man-days.

Equate: $40n = 30(n+5) \Rightarrow 40n = 30n + 150 \Rightarrow 10n = 150 \Rightarrow n = 15$.

✓ **Answer: Option (d) 15 engineers**

Q13. If 12 men and 16 boys can do a piece of work in 5 days and 13 men and 24 boys can do it in 4 days, compare the daily work done by a man with that done by a boy? (a) 1 : 2 (b) 1 : 3 (c) 2 : 1 (d) 3 : 1

■ **TRICK / SHORTCUT:** Total work is the same in both cases, so equate: (workers $\times$ days) done by group 1 = group 2. Let one man's 1-day work = m and one boy's = b , then solve the two linear equations.

Step-by-step Solution:

Total work = $5 \times (12m + 16b) = 4 \times (13m + 24b)$, where m = one man's 1-day work, b = one boy's 1-day work.

$$60m + 80b = 52m + 96b$$

$$60m - 52m = 96b - 80b \Rightarrow 8m = 16b \Rightarrow m = 2b, \text{ i.e. } m : b = 2 : 1.$$

✓ **Answer: Option (c) 2 : 1**

Q14. A can do a work in 10 days and B can do the same work in 20 days. They work together for 5 days and then A goes away. In how many more days will B finish the work? (a) 5 days (b) 6.5 days (c) 10 days (d) 8.33 days

■ **TRICK / SHORTCUT:** Same 'partway leaver' method as Q4/Q7/Q8.

Step-by-step Solution:

Combined 1 day work = $\frac{1}{10} + \frac{1}{20} = \frac{2}{20} + \frac{1}{20} = \frac{3}{20}$.

Work done in 5 days = $5 \times \frac{3}{20} = \frac{15}{20} = \frac{3}{4}$.

Remaining work = $1 - \frac{3}{4} = \frac{1}{4}$.

B alone needs $(\frac{1}{4}) \div (\frac{1}{20}) = (\frac{1}{4}) \times 20 = \mathbf{5 \text{ days}}$.

✓ **Answer: Option (a) 5 days**

Q15. 30 men working 5 h a day can do a work in 16 days. In how many days will 20 men working 6 h a day do the same work? (a) 22.5 days (b) 20 days (c) 21 days (d) None of these

■ **TRICK / SHORTCUT:** Extended man-days rule: $(\text{Men}) \times (\text{Hours/day}) \times (\text{Days}) = \text{constant total work (man-hours)}$.

Step-by-step Solution:

Total work = $30 \times 5 \times 16 = 2400$ man-hours.

For 20 men at 6 h/day: $20 \times 6 \times d = 2400 \Rightarrow 120d = 2400 \Rightarrow d = \mathbf{20 \text{ days}}$.

✓ **Answer: Option (b) 20 days**

Q16. Pipe A and B running together can fill a cistern in 6 minutes. If B takes 5 minutes more than A to fill the cistern, then find the time in which A and B will fill the cistern separately.

(a) 15 min, 20 min (b) 15 min, 10 min (c) 10 min, 15 min (d) 25 min, 20 min

■ **TRICK / SHORTCUT:** Let A's time = x , form a quadratic using $1/x + 1/(x+5) = 1/6$, and solve. Always reject the negative root.

Step-by-step Solution:

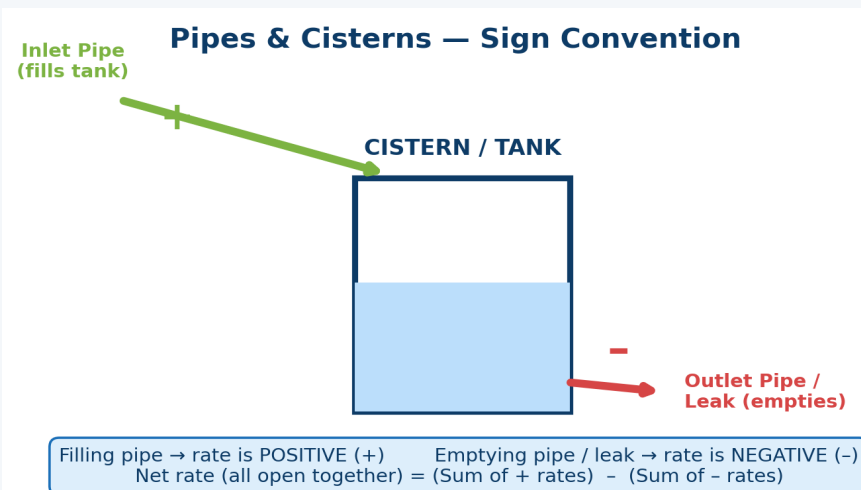
Let A alone take x minutes, so B alone takes $(x + 5)$ minutes.

$$1/x + 1/(x+5) = 1/6$$

Multiplying out: $6(x+5) + 6x = x(x+5) \Rightarrow 12x + 30 = x^2 + 5x$

$$x^2 - 7x - 30 = 0 \Rightarrow (x - 10)(x + 3) = 0 \Rightarrow x = 10 \text{ (rejecting } x = -3, \text{ time can't be negative).}$$

So A takes **10 minutes** and B takes $10 + 5 = 15$ minutes.



✓ **Answer: Option (c) 10 min, 15 min**

Q17. A takes 5 days more than B to do a certain job and 9 days more than C; A and B together can do the job in the same time as C. How many days would A take to do it?

(a) 16 days (b) 10 days (c) 15 days (d) 20 days

■ **TRICK / SHORTCUT:** Assign one variable to the fastest worker ($C = x$), express A and B in terms of x using the given differences, then use "A and B together = C alone" to form a solvable equation.

Step-by-step Solution:

Let C alone take x days. Then $A = x + 9$ days, and $B = A - 5 = x + 4$ days.

Given: A and B together take the same time as C alone, so $1/(x+9) + 1/(x+4) = 1/x$.

Cross-multiplying: $x(x+4) + x(x+9) = (x+9)(x+4)$

$$x^2 + 4x + x^2 + 9x = x^2 + 13x + 36 \Rightarrow 2x^2 + 13x + 36 = x^2 + 13x + 36 \Rightarrow x^2 = 36 \Rightarrow x = 6.$$

So C = 6 days, and $A = x + 9 = 15$ days.

✓ **Answer: Option (c) 15 days**

Q18. A cistern is normally filled in 6 h but takes 4 h longer to fill because of a leak in its bottom. If the cistern is full, the leak will empty it in how much time?

(a) 15 h (b) 16 h (c) 20 h (d) None of these

■ **TRICK / SHORTCUT:** Use the sign convention: with the leak, the NET rate = (filling rate) – (leak rate). Find the leak's rate by subtraction, then invert it for the emptying time.

Step-by-step Solution:

Normal filling rate = $1/6$ per hour. With the leak, time becomes $6 + 4 = 10$ h, so net rate = $1/10$ per hour.

Leak's emptying rate = (filling rate) – (net rate) = $1/6 - 1/10 = 5/30 - 3/30 = 2/30 = 1/15$.

So the leak alone would empty the full cistern in **15 hours**.

✓ **Answer: Option (a) 15 hours**

Q19. There are two pipes in a tank. Pipe A is for filling the tank and Pipe B is for emptying the tank. If A can fill the tank in 10 hours and B can empty the tank in 15 hours then find how many hours will it take to completely fill a half empty tank?

(a) 30 hours (b) 15 hours (c) 20 hours (d) 33.33 hours

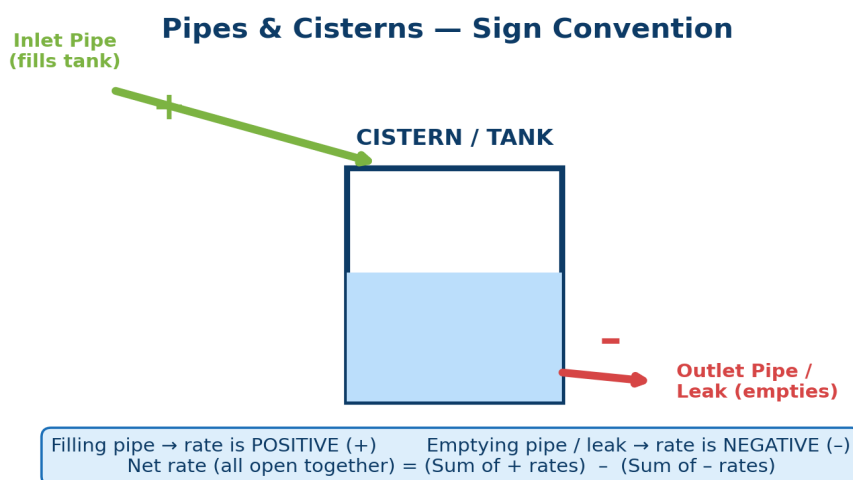
■ **TRICK / SHORTCUT:** Find the NET rate first (filling rate minus emptying rate). Since the tank is already half full, you only need to fill the remaining half, not the whole tank.

Step-by-step Solution:

Net rate (both open) = $1/10 - 1/15 = 3/30 - 2/30 = 1/30$ per hour (positive, so tank fills slowly).

Tank is already half full, so remaining work = $1/2$.

Time = $(1/2) \div (1/30) = (1/2) \times 30 = 15$ hours.



✓ **Answer: Option (b) 15 hours**

Q20. Apurva can do a piece of work in 12 days. Apurva and Amit complete the work together and were paid Rs. 54 and Rs. 81 respectively. How many days must they have taken to complete the work together?

(a) 4 days (b) 4.5 days (c) 4.8 days (d) 5 days

■ **TRICK / SHORTCUT:** Golden rule of wages: money earned is always split in the same ratio as the work done, which equals the ratio of efficiencies (since both worked the same number of days). Use the wage ratio to get Amit's efficiency, then combine.

Step-by-step Solution:

Wage ratio Apurva : Amit = 54 : 81 = 2 : 3, so efficiency ratio Apurva : Amit = 2 : 3 (they worked equal days, so pay is proportional to work/efficiency).

Apurva's efficiency = $1/12$ work/day (given). Since Amit's efficiency is $(3/2)$ times Apurva's: Amit's efficiency = $(3/2) \times 1/12 = 1/8$ work/day.

Combined efficiency = $1/12 + 1/8 = 2/24 + 3/24 = 5/24$ per day.

Days together = $24/5 = 4.8$ days.

✓ **Answer: Option (c) 4.8 days**

Q21. There are three Taps A, B and C in a tank. They can fill the tank in 10 hrs, 20 hrs and 25 hrs, respectively. At first, all of them are opened simultaneously. Then after 2 hours, tap C is closed and A and B are kept running. After the 4th hour, tap B is also closed. The remaining work is done by Tap A alone. Find the percentage of the work done by Tap A by itself.

(a) 32% (b) 52% (c) 75% (d) None of these

■ **TRICK / SHORTCUT:** Break the timeline into phases (exactly like a Gantt chart) and calculate the work done in each phase separately using LCM = 100 for clean percentage values.

Step-by-step Solution:

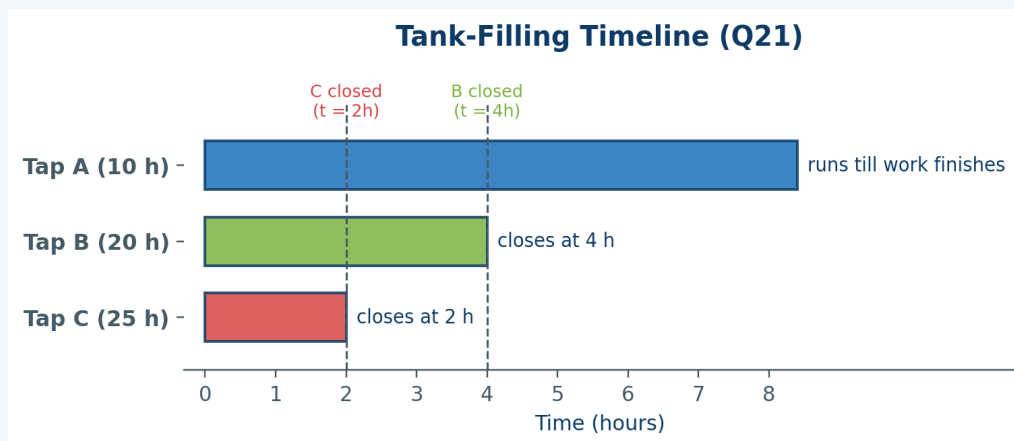
Take total work = LCM(10,20,25) = 100 units. Efficiencies: A = 10/hr, B = 5/hr, C = 4/hr.

Phase 1 (0-2 h, A+B+C open): rate = 10+5+4 = 19/hr $\Rightarrow$ work done = $2 \times 19 = 38$ units.

Phase 2 (2-4 h, only A+B open): rate = 10+5 = 15/hr $\Rightarrow$ work done = $2 \times 15 = 30$ units.

Work done up to 4th hour = 38 + 30 = 68 units. Remaining work = 100 – 68 = 32 units.

This remaining **32 units** is completed by Tap A alone, i.e. Tap A alone does 32% of the total work.



✓ **Answer: Option (a) 32%**

Q22. A cistern has two pipes. One can fill it with water in 8 hours and other can empty it in 5 hours. In how many hours will the cistern be emptied if both the pipes are opened together when 3/4 of the cistern is already full of water?

(a) 30 hours (b) 15 hours (c) 20 hours (d) 10 hours

■ **TRICK / SHORTCUT:** Find the net rate with correct signs: filling is (+), emptying is (–). If the net rate is negative, the tank empties, so just divide the current water level by the (positive) magnitude of the net emptying rate.

Step-by-step Solution:

Net rate = (filling rate) – (emptying rate) = $1/8 - 1/5 = 5/40 - 8/40 = -3/40$ per hour.

The negative sign shows the tank is net emptying, at a rate of 3/40 of the tank per hour.

Water currently present = 3/4 of the tank. Time to empty = $(3/4) \div (3/40) = (3/4) \times (40/3) = 10$ hours.

✓ **Answer: Option (d) 10 hours**

Quick Answer Key

Q.No	Answer	Q.No	Answer
1	6 days (Opt 3)	12	15 engineers (Opt d)
2	36 days (Opt 3)	13	2:1 (Opt c)
3	12 days (Opt 2)	14	5 days (Opt a)
4	6.25 days (Opt 3)	15	20 days (Opt b)
5	60 days (Opt 4)	16	10 min, 15 min (Opt c)
6	6.25 days (Opt 2)	17	15 days (Opt c)
7	3 days (Opt 1)	18	15 h (Opt a)
8	≈ 6.67 days (nearest Opt 2)	19	15 h (Opt b)
9	1.2 days (Opt 2)	20	4.8 days (Opt c)
10	Data inconsistent — verify source	21	32% (Opt a)
11	42 days (Opt d)	22	10 h (Opt d)

Revision Checklist for Students

- Memorise the two-person shortcut: Together = $ab \div (a+b)$.
- For 3+ people/pipes, use the LCM method to avoid fractions.
- 'Someone leaves partway' → always compute work already done first, then the remainder.
- In pipes & cisterns, filling = (+), emptying/leak = (–). Add efficiencies with sign.
- Wages/payment are always split in the ratio of work done (= ratio of efficiencies, if same days).